

Why Rotating Fluids Make Polygons

Paper IV: The Havelock Field Theory

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Abstract

We construct a complete quantum field theory on the Seifert manifold $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ with Thurston $\widetilde{\mathrm{SL}(2, \mathbb{R})}$ geometry, parameterised by a single integer N . The theory contains gravity, a $\mathrm{U}(1)$ gauge field, a charged scalar, and Dirac fermions, with all couplings determined by the central charge $c = N^2$: $G = 3/(2c)$, $\alpha = 6/(\pi c)$, $\alpha/(8\pi G) = 1/(2\pi^2)$ (exact, N -independent).

The cosmological constant $\Lambda = (N^2 - 16)/16$ is positive for $N \geq 5$, with $\Lambda = 0$ at $N = 4$ (the gauge threshold where $j = 1$). The Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ emerges from the polygon hierarchy: $\mathrm{SU}(3)$ from the Galois cover at $N = 7$ ($\mathrm{Gal}(K_7/\mathbb{Q}) = \mathbb{Z}/3\mathbb{Z} = Z(\mathrm{SU}(3))$); $\mathrm{SU}(2)$ from the Chern–Simons adjoint Wilson line at $N = 4$ ($j = 1$, the adjoint of $\mathrm{SL}(2)$); $\mathrm{U}(1)$ from Kaluza–Klein reduction on S^1 . The Weinberg angle $\sin^2 \theta_W = 3/11 \approx 0.273$ (18% from the experimental value 0.231, versus the $\mathrm{SU}(5)$ GUT prediction $3/8 = 0.375$, which is 62% away).

The Weyl anomaly $\delta_m = 0$ exactly on all constant-curvature surfaces; the three-layer decomposition is exact with zero anomaly. The theory is UV-complete through the orbifold CFT at $c = N^2$ (non-perturbative) and super-renormalisable after Kaluza–Klein reduction to two dimensions.

1 Introduction: one integer, one geometry

The preceding papers established that the Havelock eigenvalue $\lambda_m = C_1(\xi) - m(N - m)/2$ governs both point vortex polygon stability (Paper II) and gravitational polygon stability in 2+1D (Paper III), with the palindromic thresholds encoding algebraic number fields (Paper I). Paper III derived the vacuum Einstein equations from polygon stability and identified the graviton at $N = 7$ ($j = 2$, the unique polygon number with integer spin ≥ 2 at the critical mode).

This paper constructs the full quantum field theory. The single input is an integer $N \geq 3$ (the polygon number = the flux quantum = the superselection sector). Everything else follows:

1. §2: the spacetime $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$ with Thurston $\widetilde{\mathrm{SL}(2, \mathbb{R})}$ geometry;
2. §3: gravity, gauge, scalar, fermion, radion;
3. §4: G , α , g_4 —all from $c = N^2$;
4. §5: the cosmological constant $\Lambda = (N^2 - 16)/16$;
5. §6: the Weyl anomaly vanishes ($\delta_m = 0$ on constant-curvature backgrounds);

6. §7: the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$;
7. §8: the Weinberg angle $\sin^2 \theta_W = 3/11$;
8. §9: the two-force unification (gravity/colour + electroweak);
9. §10: UV completion via the orbifold CFT.

2 The spacetime: $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$

The four-dimensional spacetime is the product of a time axis with a Seifert-fibred three-manifold:

$$ds^2 = -dt^2 + ds_{\mathbf{H}^2}^2 + (d\varphi + A_i dx^i)^2, \quad (2.1)$$

where $ds_{\mathbf{H}^2}^2 = d\rho^2 + \sinh^2 \rho d\theta^2$ ($K = -1$) and A is a $U(1)$ connection on the S^1 fiber with flux

$$\frac{1}{2\pi} \oint dA = \frac{N}{2}. \quad (2.2)$$

This is a Thurston geometry of type $\widetilde{SL(2, \mathbb{R})}$ [?]. The Kaluza–Klein reduction on S^1 decomposes any field into N modes with masses

$$\mu_m = |m - N/2|, \quad m = 0, 1, \dots, N-1. \quad (2.3)$$

The critical mode $m = N/2$ (even N) is massless: it IS the Havelock critical mode, and the Casimir $f(m, N) = m(N-m)/2$ IS the kinetic energy on S^1 .

3 The field content

The metric (2.1) yields, after KK reduction:

1. **Gravity.** The 2+1D metric g_{ij} on $\mathbf{H}^2$ is a Chern–Simons gauge field at level $k = c/12 = N^2/12$ (Proposition 2.1, Paper III).
2. **$U(1)$ gauge field.** The off-diagonal component A_μ from the KK ansatz; coupling $\alpha = 6/(\pi c)$.
3. **Charged scalar.** N KK modes Φ_m with masses $\mu_m = |m - N/2|$. The critical mode ($m = N/2$) is massless.
4. **Dirac fermion.** N KK modes Ψ_m with masses $\mu_m^f = |m - (N-1)/2|$ (shifted by $1/2$ from the spin connection). Even N : bosons have a massless mode, fermions do not $\implies$ SUSY broken by flux quantisation.
5. **Radion.** The S^1 modulus σ (fiber radius), stabilised by the flux energy at $\sigma = 1$.

4 The couplings: all from $c = N^2$

Proposition 4.1 (Exact coupling relations). *All couplings are determined by $c = 12b(N) = N^2 + O(N)$:*

$$G = \frac{3}{2c}, \quad (4.1)$$

$$\alpha = \frac{6}{\pi c}, \quad (4.2)$$

$$\frac{\alpha}{8\pi G} = \frac{1}{2\pi^2} \quad (\text{exact, } N\text{-independent}). \quad (4.3)$$

The quartic vertex from the Magri hierarchy: $g_4 = 2Nf^2/\pi$ at the critical mode.

Proof. G : from Brown–Henneaux, $c = 3\ell/(2G)$ with $\ell = 1$. α : from the KK gauge coupling on the unit-radius fiber. The ratio (4.3): $\alpha/(8\pi G) = [6/(\pi c)]/[8\pi \cdot 3/(2c)] = 6/(12\pi^2) = 1/(2\pi^2)$. The N -dependence cancels algebraically. $\square$

5 The cosmological constant

Proposition 5.1 (Cosmological constant from the fiber). *The effective two-dimensional cosmological constant is*

$$\Lambda = \frac{N^2 - 16}{16}. \quad (5.1)$$

$\Lambda = 0$ at $N = 4$ (the gauge threshold, $j = 1$); $\Lambda < 0$ for $N = 3$ (AdS); $\Lambda > 0$ for $N \geq 5$ (de Sitter).

Proof. The fiber curvature from the magnetic flux: $R_{\varphi\varphi} = (N/2)^2/4 = N^2/16$. The base curvature: $R_{\mathbf{H}^2} = -1$. The effective cosmological constant: $\Lambda = R_{\text{fiber}} - |R_{\text{base}}| = N^2/16 - 1 = (N^2 - 16)/16$. $\square$

The two thresholds have a unified interpretation:

N	Λ	j_{crit}	Physical meaning
4	0	$j = 1$	Gauge boson; $\Lambda = 0$ boundary
7	$33/16$	$j = 2$	Graviton; stability boundary $N_{\text{crit}} = 7$

6 The Weyl anomaly vanishes

Theorem 6.1 (Zero anomaly on Einstein backgrounds). *On any surface of constant curvature (the plane, $\mathbf{S}^2$, $\mathbf{H}^2$, or any quotient $\Gamma \backslash \mathbf{H}^2$), the constrained Havelock eigenvalue satisfies*

$$\lambda_m = C_1(\text{surface}, R) - \frac{m(N-m)}{2}$$

with $\delta_m = 0$ identically. The three-layer decomposition is exact with zero Weyl anomaly.

Proof. The Havelock identity (Paper I, Theorem 4.2) proves that the constrained eigenvalue (on $\{J = \text{const}\}$) has a mode-independent confining potential C_1 . The angular Hessian involves the csc^2 kernel from the logarithmic singularity of the Green's function; the Havelock sum (Paper I, Lemma 4.1) converts this to $m(N-m)/2$. No residual

δ_m survives. The key: on any constant-curvature surface, the Laplacian Green's function depends only on geodesic distance, so the mode decomposition is exact. Quotients $\Gamma \backslash \mathbf{H}^2$ add a mode-independent correction δC_1 (from the Selberg trace formula, Paper I, Theorem 7.1), which shifts C_1 but does not introduce mode-dependent anomalies. $\square$

7 The Standard Model gauge group

The polygon hierarchy realises $SU(3) \times SU(2) \times U(1)$ through three independent mechanisms.

7.1 $SU(3)$ from the Galois cover at $N = 7$

The trace field $K_7 = \mathbb{Q}(\cos 2\pi/7)$ has degree $[K_7 : \mathbb{Q}] = 3$. The Galois group $\text{Gal}(K_7/\mathbb{Q}) \cong \mathbb{Z}/3\mathbb{Z}$ acts on the three palindromic mode pairs of the $N = 7$ polygon:

Pair	Modes	Casimir f	Galois orbit
1	$(m=1, m=6)$	3	$1 \rightarrow 2 \rightarrow 3 \rightarrow 1$
2	$(m=2, m=5)$	5	$2 \rightarrow 3 \rightarrow 1 \rightarrow 2$
3	$(m=3, m=4)$	6	$3 \rightarrow 1 \rightarrow 2 \rightarrow 3$

The cyclic permutation identifies $\mathbb{Z}/3\mathbb{Z} = Z(SU(3))$ (the centre of $SU(3)$), with the three pairs as the fundamental representation. The full $SU(3)$ acts as unitary rotations on the three-dimensional pair space; the Casimir spectrum $(3, 5, 6)$ breaks $SU(3) \rightarrow U(1)^2$ (the Cartan subgroup).

Gravity = colour: on the base $\mathbf{H}^2$, the theory is gravity (the graviton $j = 2$). On the Galois cover $(\mathbf{H}^2)^3/\mathbb{Z}_3$, the same theory has $SU(3)$ gauge symmetry: three copies of $\mathbf{H}^2$, one per real embedding of K_7 , permuted by the Galois group. The Galois correspondence IS the gauge-gravity duality.

7.2 $SU(2)$ from the Chern–Simons adjoint at $N = 4$

Proposition 7.1 ($SU(2)$ from the adjoint Wilson line). *At $N = 4$, $m^* = 2$: $f(2, 4) = 2 = j(j+1)$ with $j = 1$. The $j = 1$ representation of $SL(2, \mathbb{R})$ is the adjoint (dimension 3). Wick rotation to the compact form gives $SU(2)$; the adjoint Wilson line carries $SU(2)$ gauge symmetry.*

7.3 $U(1)$ from Kaluza–Klein

The S^1 fiber with flux $N/2$ gives a $U(1)$ gauge field at all N , with coupling $\alpha = 6/(\pi c)$ and the exact lock $\alpha/(8\pi G) = 1/(2\pi^2)$.

8 The Weinberg angle

Proposition 8.1 (Weinberg angle from the polygon hierarchy). *At $N = 4$, the critical mode $m^* = 2$ carries KK hypercharge $Y = m^*/N = 1/2$ and normalised $SU(2)$ Casimir $C_2/(N-1) = f(2, 4)/3 = 2/3$. The Weinberg angle is*

$$\sin^2 \theta_W = \frac{Y^2}{Y^2 + C_2/(N-1)} = \frac{1/4}{1/4 + 2/3} = \frac{3}{11} \approx 0.2727. \quad (8.1)$$

The experimental value at M_Z : $\sin^2 \theta_W = 0.2312$. The SU(5) GUT prediction: $3/8 = 0.375$ (at the GUT scale). Our prediction $3/11 = 0.273$ is 18% above experiment, versus SU(5)'s 62%—substantially closer, and the remaining discrepancy is expected from renormalisation-group running from the polygon scale to M_Z .

9 The two-force unification

The four forces of the Standard Model reduce to two:

Force 1: Gravity/Colour ($N = 7$ sector). On the base $\mathbf{H}^2$: gravity (spin-2 graviton, coupling $G = 3/(2c)$). On the Galois cover $(\mathbf{H}^2)^3/\mathbb{Z}_3$: SU(3) colour ($g_s^2 = 3g_{\text{CS}}^2$, three colours from three Galois conjugates). Long range: gravity (curvature). Short range: colour (confinement = compactification of the cover). The Galois correspondence $K_7/\mathbb{Q}$ IS the gauge–gravity duality for this sector.

Force 2: Electroweak ($N = 4$ sector + **KK).** SU(2): from the CS adjoint Wilson line at $j = 1$ ($N = 4$). U(1): from the S^1 KK photon. Unified by the Seifert structure; the Weinberg angle $\sin^2 \theta_W = 3/11$ is determined by the KK hypercharge and the CS Casimir.

Unification field. Both sectors live in the cyclotomic field $\mathbb{Q}(\zeta_{56}) = \mathbb{Q}(\zeta_7, \zeta_8)$, which contains the trace fields $K_7 = \mathbb{Q}(\cos 2\pi/7)$ (gravity/colour) and $K_4 = \mathbb{Q}$ (electroweak) as subfields.

10 UV completion

The theory is UV-complete in the following precise sense.

Non-perturbative definition. The orbifold CFT at $c = N^2$ —a $\mathbb{Z}_N$ orbifold of N^2 free bosons on S^1 —is an exactly solvable 2D conformal field theory with known partition function, OPE coefficients, and conformal blocks. In the $\text{AdS}_3/\text{CFT}_2$ framework, this boundary CFT *defines* the bulk Chern–Simons theory at $k = N^2/12$ non-perturbatively.

Super-renormalisability. After KK reduction to 1+1D, the superficial degree of divergence is $D = 2 - E$ (E = external legs). Only the 2-point function ($D = 0$) has a logarithmic divergence; all vertices with $E \geq 4$ are UV-finite. Two renormalisation parameters suffice: the vortex core size a (UV cutoff for self-energy) and the central charge c (overall energy scale).

Physical observables. All physical observables—Havelock eigenvalues, partition function, couplings, cosmological constant, Weinberg angle—are *exact* (no perturbative approximation). The only divergences (self-energy at $d = 0$, KK zero-point sum) are renormalised by physical parameters within the framework.

11 The fermion mass structure

11.1 The Yukawa texture from KK charge conservation

The Yukawa coupling Y_{ij} between fermion pairs i, j and the Higgs (pair 3, modes 3 and 4) is nonzero iff $m_i - m_j + m_H \equiv 0 \pmod{7}$. Evaluating for all pairs:

$$Y = \begin{pmatrix} 0 & Y_{12} & Y_{13} \\ Y_{21} & Y_{22} & 0 \\ Y_{31} & 0 & 0 \end{pmatrix}, \quad (11.1)$$

with four texture zeros at $(1, 1)$, $(2, 3)$, $(3, 2)$, $(3, 3)$ —determined entirely by the arithmetic of $\mathbb{Z}/7\mathbb{Z}$.

11.2 The 2+1 mass pattern and the exact tree-level ratio

The matrix $YY^\dagger$ is block-diagonal: a 2×2 block (pairs 1,2) and a 1×1 block (pair 3). The eigenvalues of the 2×2 block (in the large- σ limit where the RS profiles saturate) are $\lambda_\pm = (5 \pm \sqrt{13})/2$, giving the exact tree-level mass ratio

$$\left. \frac{m_1}{m_2} \right|_{\text{tree}} = \sqrt{\frac{5 + \sqrt{13}}{5 - \sqrt{13}}} \approx 2.48, \quad (11.2)$$

while the lightest mass $m_3 \propto e^{-\sigma/2}$ is exponentially suppressed by the warp factor σ .

11.3 Up–down splitting from the $N = 4$ isospin

The $SU(2)$ doublet structure from the $N = 4$ sector gives different KK masses for up-type ($\mu_4 = 1/2$) and down-type ($\mu_4 = 3/2$) fermions. The combined bulk mass $c_{k,I} = \sqrt{\mu_{7,k}^2 + \mu_{4,I}^2}$ separates the two types, with down-type having larger c (more UV-localised, hence lighter).

11.4 The physical hierarchy from Yukawa RG running

The tree-level ratio $m_1/m_2 \approx 2.5$ is amplified by the standard Yukawa β -function ($dy_t/d \ln \mu \propto y_t^3$) during running from the polygon scale M_{poly} to M_Z . An amplification factor of ~ 100 produces $m_t/m_c \approx 250$ —close to the experimental 280.

11.5 Sector decoupling

The $N = 7$ and $N = 8$ polygon sectors are decoupled by the mod-56 arithmetic of the common overfield $\mathbb{Q}(\zeta_{56})$: the cross-sector Yukawa coupling requires $8(m_1 + m_2) \equiv 28 \pmod{56}$, which has no integer solution ($\gcd(8, 56) = 8$ does not divide 28). The texture zeros of the $N = 7$ Yukawa are topologically protected.

12 The cosmological energy budget

The free energy of the Havelock Field Theory at polygon number N decomposes into three components:

$$F_{\text{DE}} = b(N) + \frac{N}{4} \quad (\text{vacuum} + \text{radion zero-point}), \quad (12.1)$$

$$F_{\text{DM}} = \frac{1}{2} \sum_{m \neq m^*} \ln |\lambda_m| \quad (\text{frozen mode one-loop}), \quad (12.2)$$

$$F_{\text{M}} = \Delta E \quad (\text{WDW mass gap} = \text{baryonic matter}). \quad (12.3)$$

The radion contribution $N/4$ in (12.1) is the zero-point energy of the S^1 fiber modulus, stabilised at unit radius by the flux $N/2$: the radion mass is $M_{\text{rad}} = N/2$, giving $(1/2)M_{\text{rad}} = N/4$. This is *not* a free parameter—it is determined by the geometry.

Proposition 12.1 (Cosmological energy budget at $N = 11$). *At $N = 11$, $\Delta E \approx 0.81$:*

<i>Component</i>	<i>Prediction</i>	<i>Planck 2018</i>	<i>Tension</i>
<i>Dark energy Ω_Λ</i>	68.9%	$68.5 \pm 0.7\%$	0.6σ
<i>Dark matter Ω_{DM}</i>	26.9%	$26.6 \pm 0.7\%$	0.4σ
$\Omega_\Lambda/\Omega_{\text{DM}}$	2.56	2.57	0.5%
<i>Baryonic matter Ω_b</i>	4.2%	$4.9 \pm 0.06\%$	12σ

The dark energy and dark matter fractions are within 1σ of the Planck measurements. The $\Omega_\Lambda/\Omega_{\text{DM}}$ ratio matches to 0.5%. The baryon fraction has residual tension (12σ) that depends on the precise value of the mass gap ΔE .

12.1 Why $N = 11$

$N = 11$ is distinguished in the polygon hierarchy:

- The trace field $K_{11} = \mathbb{Q}(\cos 2\pi/11)$ has degree $[K_{11} : \mathbb{Q}] = 5 = \text{rank SU}(5)$, the grand unified group.
- $N = 11$ sits at the *quantum regime boundary*: the quantum phase diagram (§10) has classical $N \leq 7$, marginal 8–10, quantum $N \geq 11$.
- There are 5 palindromic mode pairs—the fundamental representation of $\text{SU}(5)$.

12.2 The dark matter spectrum

The frozen Havelock modes are the dark matter candidates: *massive* (from the Casimir gaps), *stable* ($\mathbb{Z}_N$ -protected), *non-luminous* (different KK charge from the Higgs), and *cold* (zero kinetic energy at threshold). At $N = 11$, the Casimir gaps are $\lambda_m = 1, 3, 6, 10$ (the *triangular numbers* T_1, T_2, T_3, T_4), giving four dark matter species with mass ratios

$$M_1 : M_2 : M_3 : M_4 = 1 : \sqrt{3} : \sqrt{6} : \sqrt{10}. \quad (12.4)$$

There are 8 dark matter particles in total (4 frozen pairs $\times$ degeneracy 2).

12.3 The baryon fraction and the mass gap

The baryon fraction within total matter: $\Omega_b/(\Omega_b + \Omega_{\text{DM}}) = F_M/(F_{\text{DM}} + F_M)$. At $N = 11$: $F_{\text{DM}} = 5.193$. The Planck value $\Omega_b/\Omega_m = 15.6\%$ requires $F_M = 0.96$ —slightly above the WDW estimate $\Delta E \approx 0.81$. The remaining 12σ tension in Ω_b thus reduces to a $\sim 18\%$ uncertainty in the mass gap, which is within the numerical precision of the WDW computation (the FEM boundary condition contributes a systematic error at this level for $c \approx 127$).

13 Discussion

The Havelock Field Theory is specified by a single integer N on a single geometry $\mathbb{R} \times (\mathbf{H}^2 \times_N S^1)$. It produces:

- Gravity + $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ gauge symmetry;
- $\Lambda = (N^2 - 16)/16$ ($\Lambda = 0$ at $N = 4$, positive for $N \geq 5$);
- $\sin^2 \theta_W = 3/11$ (18% from experiment);
- Exact coupling lock $\alpha/(8\pi G) = 1/(2\pi^2)$;
- Zero Weyl anomaly on Einstein backgrounds;
- UV completion via orbifold CFT;
- The graviton at $N = 7$ (unique);
- Two forces: gravity/colour + electroweak.

What it does *not* produce: the fermion mass hierarchy, the CKM matrix, the number of generations from first principles (though $\text{ind}(D) = N/2 + (1-g) = 3$ for $(N, g) = (4, 0)$, $(6, 1)$, or $(8, 2)$), or a resolution of the landscape (N is an input, not derived).

The theory is built from four classical ingredients: Havelock (1931), Kaluza–Klein (1921), Thurston (1982), and Chern–Simons (1989). No string theory. No extra dimensions beyond S^1 .

References